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REVIEW

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Current status, trend and progress of stem cells in tracheal tissue engineering research: an overview and perspectives

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Abstract

Currently, the reconstruction of the tracheal structure and function following a lengthy segmental tracheal defect remains a global issue that urgently needs to be solved. Tracheal tissue engineering (TTE) brings hope for tracheal reconstruction, and stem cells have shown great potential in constructing ideal tissue-engineered tracheae with their strong self-renewal ability, multidirectional differentiation potential, paracrine performance, and immunomodulatory properties. By conducting bibliometric and visual analysis, this review investigates the main countries, institutions, and authors in the field of TTE stem cell research, as well as analyzes the international research status, trends, and hotspots in this field, and reveals the possible research directions in the future. Furthermore, in this review, we summarize the types of stem cells that have been applied in TTE studies and the construction strategies for stem cell-based tissue-engineered trachea. And the current challenges of constructing a tissue-engineered trachea based on stem cells are also delved into in this review.

Keywords Stem cells, Trachea, Reconstruction, Tissue engineering, Bibliometrics

Background

Presently, clinical tracheal reconstruction is a difficult task to tackle around the world. For long tracheal defects (50% or greater in adults or 30% or more in children) due to tumor invasion, tracheal trauma, tracheomalacia, infection, etc., replacement of the trachea is advised [1–3]. The ideal trachea replacement, with suitable biomechanical properties, excellent biocompatibility, no immunogenicity, and no inflammatory response, is urgently needed to reconstruct the trachea both anatomically and functionally [4–6].

Over the last few years, tissue engineering has seen tremendous growth, leading to positive results in areas such as trachea reconstruction [3, 7–10], nerve regeneration [11–14], vascular reconstruction [15–18], cartilage repair [19–22], and bone reconstruction [23–26], and hinting at its boundless potential. Seed cells, as one of the three elements of tracheal tissue engineering (TTE), are a critical factor in the successful fabrication of a tissue-engineered trachea [10, 27]. Stem cells have been widely praised by TTE researchers due to their multifaceted differentiation potential, immunomodulatory properties, and paracrine features [28, 29]. In recent years, with the continuous

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advancement of stem cell research, numerous scientists have employed stem cells in research related to TTE. By utilizing various types of stem cells, such as mesenchymal stem cells (MSCs) and pluripotent stem cells, as well as stem cell-derived exosomes, researchers have developed a variety of tissue-engineered trachea grafts that have demonstrated positive effects and have the great potential to be ideal tracheal substitutes [30–34].

Bibliometric analysis is an effective method for qualitative and quantitative analysis of the impact of publications in the field of science, which can be conducted to investigate the research status, trends, and emerging hotpots through dynamic analysis of publications spatially and temporally [35–37]. In addition, it can also extract the authors, institutions, countries, and other information of the publications to objectively reveal the cooperative relationship between the authors, institutions, and countries, and evaluate the academic contribution of them in the research field [38, 39].

This review has carried out a comprehensive overview of stem cells in TTE. In this review, we investigate the research trend, research status, and current research hotspots of the field of stem cell research in TTE based on bibliometric analysis of relevant publications, as well as uncover the major influential countries, institutions, and authors in the research field. Furthermore, a summary of the current stem cell sources in TTE research is provided, along with an exploration of methods for constructing tissue-engineered tracheae based on stem cells and the challenges confronting this field.

Bibliometric and visualization analysis

In order to better understand the trend, current situation and hotspots of stem cells in tissue-engineered trachea research, we conducted bibliometric and visualization analysis of the related publications. The publications analyzed in this study were obtained from the Web of Science Core Collection (WOSCC). And the search strategy was TS = (stem cell* OR stem-cell*) AND TS = trachea* AND TS = (tissue engineering OR tissue engineered OR tissue-engineered) AND Publication Date = (2003–01-01 to 2024–12-31). "Articles" and "Review Articles" were the only document types, and the language was restricted to English. What's more, proceeding papers, early access, publications with expression of concern, and retracted publications were all excluded from consideration. And a total of 283 publications were finally identified and selected for bibliometric and visual analysis, as illustrated in Fig. 1.

A comprehensive collection was put together, comprising the full records and cited references of the identified publications, which included titles, authors, institutions, countries/regions, journals' names, publication year, abstracts, and keywords, and the data above was downloaded to a txt format. Furthermore, information from "Citation Report" and "Analyze Results", such as H-index, the total times cited, the annual publication amount, and the national publication count, was obtained and then managed with Microsoft Excel 2021. GraphPad 9 was use to analyze the annual publications, the annual citations, and national/regional publications. Further, to gain insight into the selected publications, other key data such as the national/regional annual publishing volume, the total frequency of citations, the average citations of each item, H-index, and the number of institutional publications, were analyzed using Origin 2021. Both VOSviewer and CiteSpace, two free software programs, are regularly utilized for bibliometric and visualization analysis. In this study, the plain text file downloaded from WOSCC was imported into VOSviewer (v. 1.6.19) and CiteSpace (v. 6.3.R1) each for analysis and knowledge graph generation.

Yearly publication trends across the globe

Stem cells have been continuously studied, bringing new hope for scholars to prepare an ideal tissue engineering trachea [40–43]. The number of global publications related to the field of stem cells in TTE rose undulatedly from 2003 to 2013, reaching a peak in 2014. Prior to 2010, the research related to stem cells in TTE was in the initial stage, with an average of 2.4 related articles published annually. And the research investment in the field got a rapid growth during the five-year span from 2010 to 2014, with an average of 15.8 publications per year, which was significantly higher than the average of the preceding seven years. A downward trend in the publication of relevant articles was shown after 2014, which then began to turn upward in 2018 and formed a second peak in 2021. Furthermore, there was a sharp increase in publication citations from 2003 to 2021, after which a fluctuation was observed. On the whole, there was a certain relationship between the cited frequency of publications and the quantity of publications. What's more, to some extent, the rise in the number of publications was accompanied by an increase in the frequency of citations, which meant that the changes in the trend of the publication volume were an important factor affecting the trends of the citations. Visualized in Fig. 2, these could be perceived.

Distribution of important contributing countries/regions, institutions, and authors

Countries/regions

The ten countries with the most total number of stem-cells related publications in tissue-engineered trachea could be seen in Fig. 3a. The United States was the most productive country, with 73 articles published. China published 54 articles, exceeding England by seven, securing the second and third spots, respectively. Furthermore,

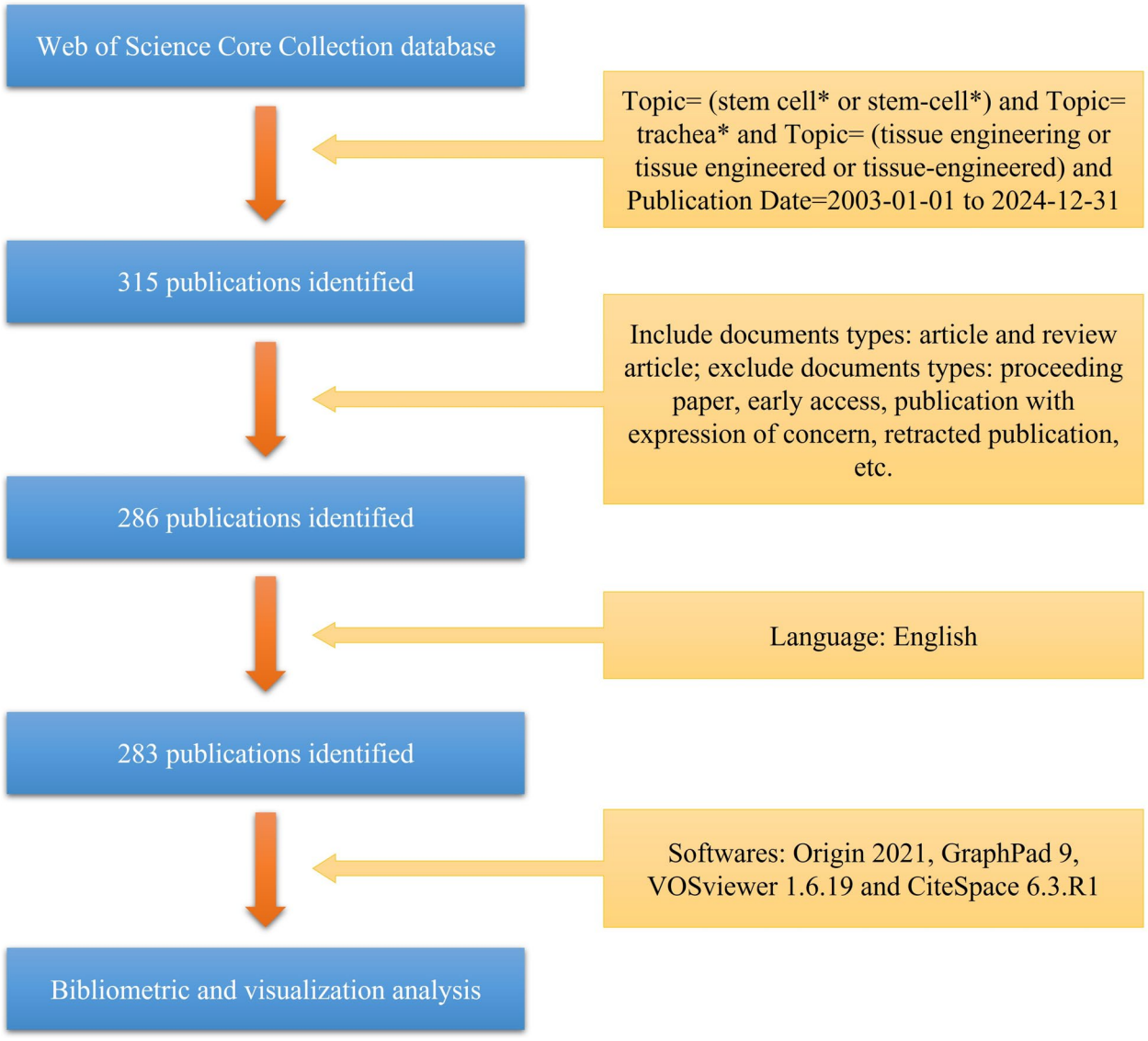


Fig. 1 Flowchart for inclusion of publications for bibliometric and visual analysis

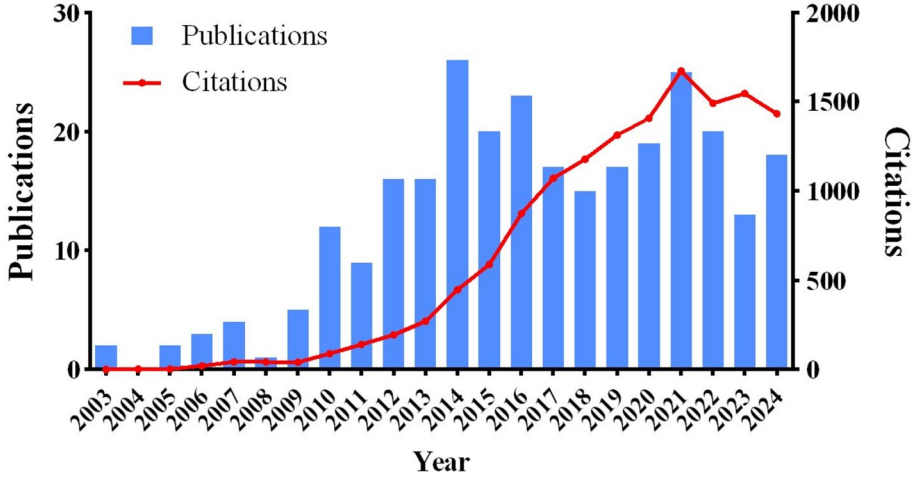


Fig. 2 Temporal progression of the number and citation frequency of publications concerning stem cells in TTE research

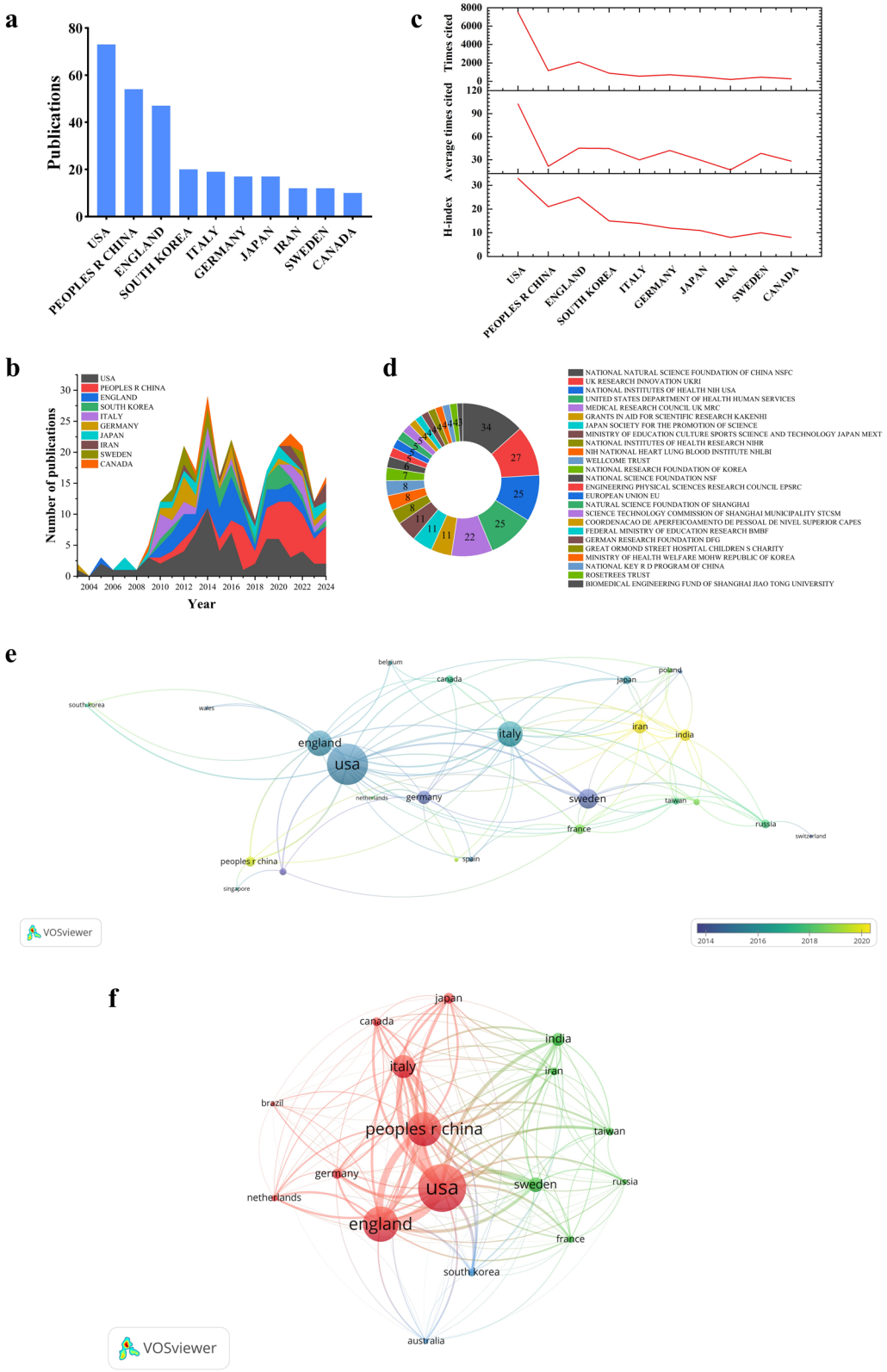


Fig. 3 The spread of the countries/regions with high contributions in the field of stem-cells research related to TTE. **a** Top 10 countries by number of publications. **b** Annual publication volume and **c** Publication quality of the top 10 countries that had published most. The top 25 **d** fundings that had funded for publishing the most papers related to TTE research with stem cells. Knowledge maps for **e** co-authorship analysis and **f** bibliographic coupling analysis of countries/regions

the combined total of publications from the top three countries constituted 61.5% of the overall number of publications analyzed. Additionally, as depicted in Fig. 3b, stem-cells research in TTE field was pioneered in the USA and Germany (2003), followed by England (2005). And relevant papers were published in the United States every year from 2003 to 2024, with the exception of 2004. China just started in 2010, but it was growing rapidly.

Evaluating the quality of a country's publications and reflecting the country's influence in a particular field of research, the H-index and citations are often taken into account [39, 44–46]. Thus, we tallied the total number of citations, the average citations per publication, and the H-index of the top 10 countries, which had been listed in Fig. 3c. The top 3 countries in terms of both total times cited and H-index were, in order, the United States, England and China. And the United States showed the most average times cited per item. It can be seen that the United States held the top spot in all three categories, indicating its high quality of the publications, with many highly-cited papers [47–49] of high quality.

Whether between countries/regions or institutions, cooperation is essential, which is conducive to the innovation and progress of scientific research [35, 50, 51]. As shown in Fig. 3e, the presence of many connections between countries suggested that international cooperation is a worldwide occurrence. The magnitude of the node reflected the degree of the total link strength (TLS) value, either high or low, with a higher TLS indicating richer cooperation. The United States led the pack in terms of TLS at 45, while England was close behind at 28, Italy at 28, Sweden at 22, Germany at 15 (same as Iran), India at 13, and China at 11 (tied with France). And the accomplishments of the United States in this area are to a certain degree connected to its early involvement in research and emphasis on international cooperation.

Besides, the knowledge map regarding the bibliographies coupling analysis of countries/regions, generated by VOSviewer, is portrayed in Fig. 3f. The coupling relationships had been widely existent between countries. And the USA was the leader in TLS (25,685), trailed by England (19,002), China (18,355), Italy (12,394), and Sweden (8,043). Moreover, thick links were apparent between the United States, England, and China in stem cells in TTE research, demonstrating tighter coupling relationships. Therefore, it was evidenced that the United States, England, and China had extensive and advanced research in the realm of stem cells in TTE research. What's more, their achievements were inseparable from the funding support, and the top 5 fundings that funded the most publications were all from them, as shown in the funding distribution graph (Fig. 3d).

Notably, in the research on using stem cells to construct tissue-engineered trachea, the United States, England, and China have slightly different research priorities. In terms of stem cell type selection, China mainly used MSCs for research, while England and the United States, in addition to using MSCs, have respectively explored the feasibility of applying airway basal stem cells (ABSCs) and induced pluripotent stem cells (iPSCs) to construct tissue-engineered tracheae [41, 52]. And research in the United States and China pays more attention to the chondrogenic differentiation of stem cells and the reconstruction of tracheal cartilage, while research in England mostly focuses on the regeneration of tracheal epithelium. In terms of tracheal scaffolds for stem cell loading, the United States, England, and China have all adopted the combination of a decellularized matrix and stem cells to construct tissue-engineered tracheae. However, among them, the United States not only focuses on exploring the feasibility of combining different types of stem cells with an acellular matrix, but also focuses on the strategy of using 3D printing technology to combine stem cells with polymer scaffolds to create personalized bionic tissue engineering tracheal scaffolds. China's research has placed emphasis on exploring suitable acellular matrix preparation protocols and the combination of natural biomaterials (such as gelatin) with stem cells. The focus of the research in England, apart from acellular matrices, also includes the introduction of nanomaterials, emphasizing the interaction between stem cells and carrier scaffolds. On the whole, the United States is the global leader in this field. Despite getting off to a late start in relevant research and international cooperation, China has now been at the forefront of the world in this field. However, its rapid development has also brought some problems, especially the pursuit of quantity of papers accompanied by the neglect of article quality, which has also caused the high total article citation frequency but the low average citation frequency in China.

Institutions

As shown in the results of the institutional analysis obtained in this study, related research on stem cells in TTE was concentrated in the institutions presented in Fig. 4. The University of London had the greatest output of publications, as depicted in Fig. 4a. The University College London (UCL) was highlighted in the institutional cooperation network (Fig. 4b) and bibliographic coupling analysis map (Fig. 4c), which implies its leading position with a heightened level of cooperation and information sharing in TTE stem-cell research. And UCL has been the source of many influential articles that promoted the development of studies in this field. One [53] of them, published in the Lancet, has been referenced over 400 times. Thus, the UK's academic achievements in this domain are inextricably linked to institutions like UCL. And China's rapid development in this field is

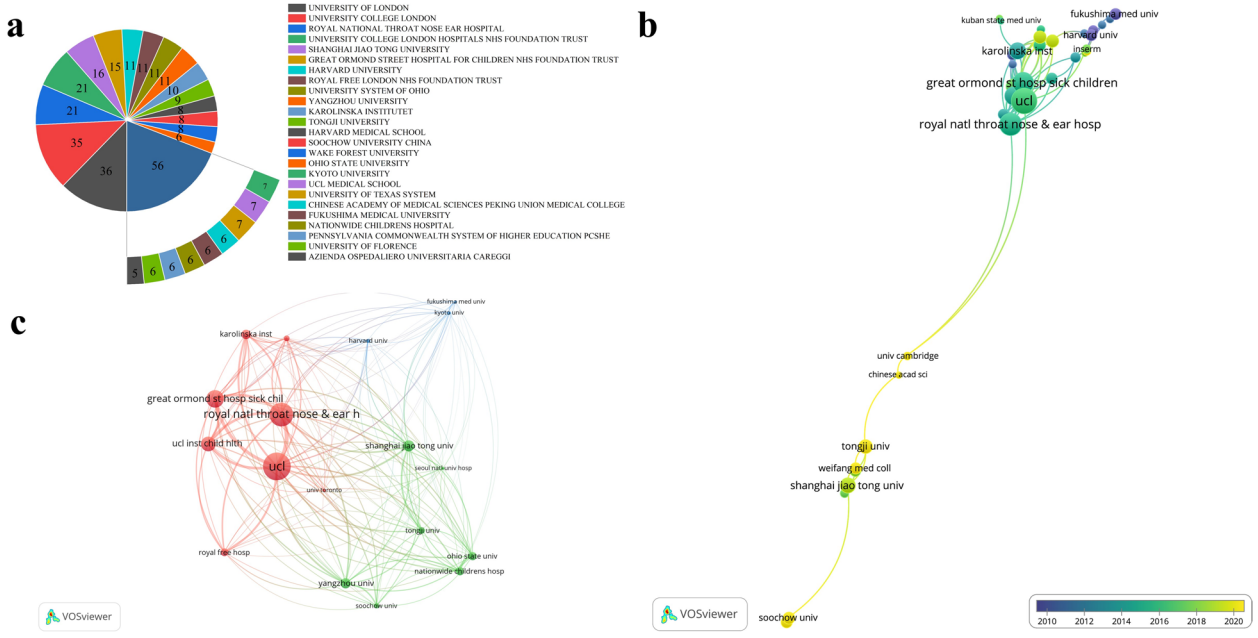


Fig. 4 Distribution of great contributive institutions related to stem cells in TTE research. **a** Showing of the top 25 institutions with the most publications. **b** Institutional cooperation network. **c** Graph of institutional bibliographic coupling analysis

inseparable from the rise of the Shanghai Jiao Tong Univ, Yangzhou Univ, Tongji Univ, and others, as shown in Fig. 4c. Furthermore, institutions have been collaborating extensively on related research on stem cells in TTE (Fig. 4b), which effectively promotes the development of such research. And those with a high level of collaboration began to do so around 2015, which coincided with the rapid growth of related publications worldwide during those years.

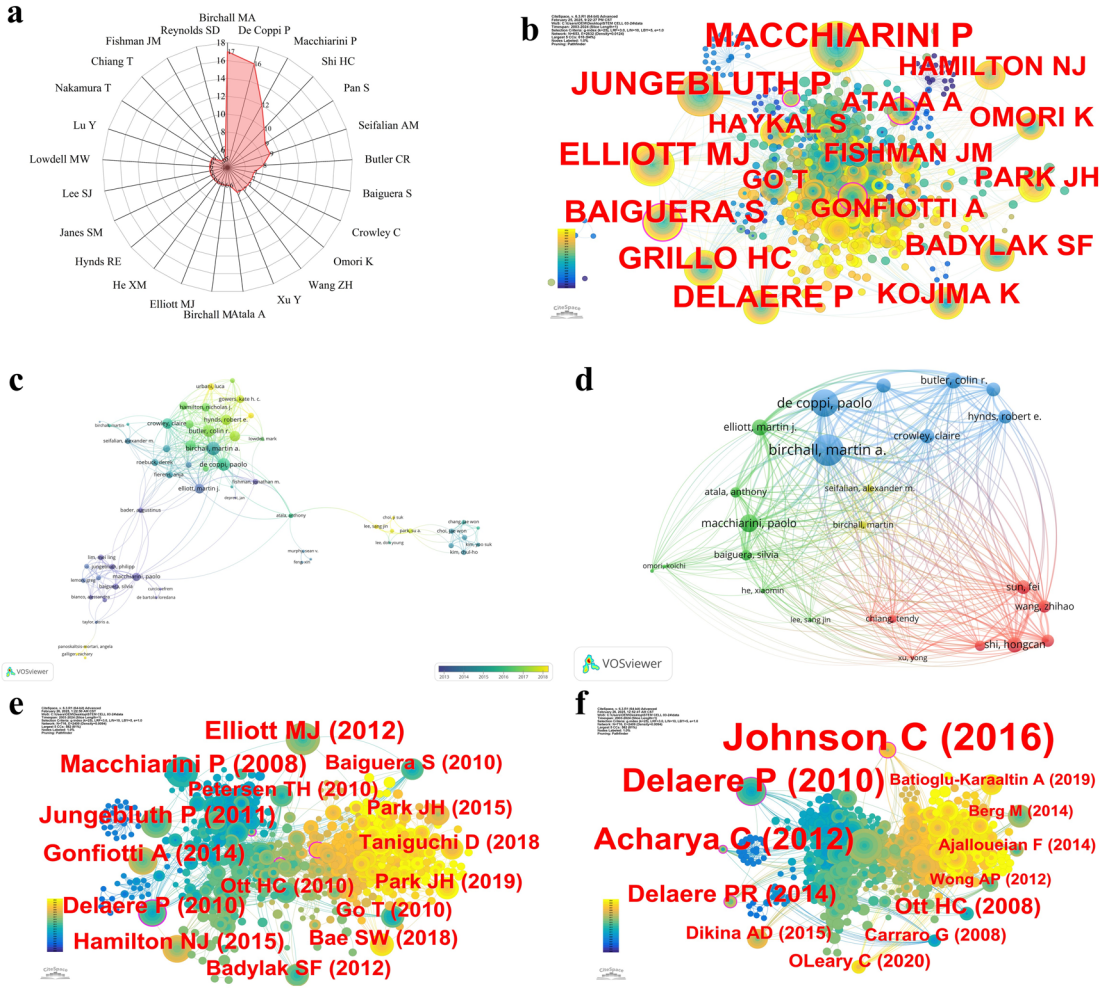
Authors

Figure 5a demonstrated that the top 25 authors in terms of published articles had a minimum of five articles each, of whom four had 10 more publications: Birchall MA (17), De Coppi P (16), Macchiarini P (12), and Shi HC (10). Besides, Macchiarini P (113 times), located at the largest node, had the highest co-citation frequency, followed by Jungebluth P (89 times), and Elliott MJ (82 times) as presented in Fig. 5b generated by CiteSpace. Not surprisingly, they all have a high level of cooperation, as evidenced by Fig. 5c. And this suggests that these scholars held an important position in the construction of tissue-engineered tracheae from stem cells. Furthermore, the authors presented in the bibliographic coupling mapping (Fig. 5d) were mainly clustered into three groups, indicating that scholars in each cluster were tackling similar or even the same research problems. Additionally, the research direction of researchers from different groups also had a broad correlation.

Cannot be ignored, Macchiarini P, who held an important position in the results of the author analysis, was revealed to have committed academic misconduct. One [54] of his articles published in the Lancet in 2008, which had a sustained citation burst over the five years from 2009 to 2013 (Fig. 5g), was eventually retracted in October 2023 due to the fabrication of experimental results that were confirmed. This may be the reason for the high co-citation frequency but low centrality of this article, as evidenced by Fig. 5e and 5f. His academic misconduct dealt a major blow to tracheal transplant research. And neither of his articles has appeared in the list of citation outbreaks in recent years (Fig. 5g).

Evaluation of journals

Journals, as the bearers of academic papers, are essential for the dissemination of research findings and the advancement of science. From 2003 to 2024, there had been in excess of 150 journals focusing on stem cells in TTE research. And Fig. 6a presented a ranking of the top 25 journals for publishing relevant papers, which together account for 37% of the total number of articles. Of these, Biomaterials had published the largest number of scholarly papers in relation to stem cells in TTE research. And according to the results of bibliographic coupling analysis, the journal Biomaterials is the most relevant one in the field of research on stem cells in TTE with the highest bibliographic coupling TLS strength (2470 TLS, Fig. 6c). Furthermore, Biomaterials had the greatest frequency of co-citations (230 times, Fig. 6b) and the strongest co-citation TLS strength (95,904 TLS, Fig. 6d), signifying its major impact in this area. The article titled "A novel tissue-engineered trachea with a mechanical behavior



g Top 25 References with the Strongest Citation Bursts

References	Year	Strength	Begin	End	2003 - 2024
Macchiarini P, 2008, LANCET, V372, P2023, DOI 10.1016/S0140-6736(08)61598-6, DOI	2008	16.64	2009	2013	
Delaere P, 2010, NEW ENGL J MED, V362, P138, DOI 10.1056/NEJMoa0810653, DOI	2010	8.06	2010	2015	
Jungebluth P, 2009, J THORAC CARDIOV SUR, V138, P586, DOI 10.1016/j.jtcvs.2008.09.085, DOI	2009	6.1	2010	2014	
Bader A, 2010, J CELL MOL MED, V14, P1877, DOI 10.1111/j.1582-4934.2010.01073.x, DOI	2010	5.28	2010	2013	
Go T, 2010, J THORAC CARDIOV SUR, V139, P437, DOI 10.1016/j.jtcvs.2009.10.002, DOI	2010	6.74	2011	2013	
Baiguera S, 2010, BIOMATERIALS, V31, P8931, DOI 10.1016/j.biomaterials.2010.08.005, DOI	2010	6.07	2011	2015	
Jungebluth P, 2011, LANCET, V378, P1997, DOI 10.1016/S0140-6736(11)61715-7, DOI	2011	10.41	2012	2016	
Petersen TH, 2010, SCIENCE, V329, P538, DOI 10.1126/science.1189345, DOI	2010	7.07	2012	2015	
Ott HC, 2010, NAT MED, V16, P927, DOI 10.1038/nm.2193, DOI	2010	7.07	2012	2015	
Jungebluth P, 2012, CLIN PHARMACOL THER, V91, P81, DOI 10.1038/clpt.2011.270, DOI	2012	5.4	2012	2015	
Elliott MJ, 2012, LANCET, V380, P994, DOI 10.1016/S0140-6736(12)60737-5, DOI	2012	13.06	2013	2017	
Badylak SF, 2012, LANCET, V379, P943, DOI 10.1016/S0140-6736(12)60073-7, DOI	2012	7.47	2013	2015	
Berg M, 2014, TISSUE ENG PT A, V20, P389, DOI 10.1089/ten.tea.2012.0514, DOI	2014	6.03	2014	2018	
Gonfiotti A, 2014, LANCET, V383, P238, DOI 10.1016/S0140-6736(13)62033-4, DOI	2014	9.44	2015	2019	
Hamilton NJ, 2015, AM J TRANSPLANT, V15, P2750, DOI 10.1111/ajt.13318, DOI	2015	10.74	2016	2020	
Park JH, 2015, BIOMATERIALS, V62, P106, DOI 10.1016/j.biomaterials.2015.05.008, DOI	2015	8.68	2017	2020	
Law JX, 2016, INT J PEDIATR OTORHI, V91, P55, DOI 10.1016/j.ijporl.2016.10.012, DOI	2016	6.07	2018	2021	
Gao MC, 2017, SCI REP-UK, V7, P0, DOI 10.1038/s41598-017-05518-3, DOI	2017	5.13	2018	2021	
Bae SW, 2018, INT J MOL SCI, V19, P0, DOI 10.3390/ijms19061624, DOI	2018	7.02	2019	2024	
Taniguchi D, 2018, INTERACT CARDIOV TH, V26, P745, DOI 10.1093/icvts/ivx444, DOI	2018	6.28	2019	2024	
Batioglu-Karaaltin A, 2019, CLIN EXP OTORHINOLAR, V12, P86, DOI 10.21053/ceo.2018.00486, DOI	2019	5.16	2019	2024	
Xu Y, 2017, ACTA BIOMATER, V58, P113, DOI 10.1016/j.actbio.2017.05.010, DOI	2017	7.46	2020	2022	
Butler CR, 2017, BIOMATERIALS, V124, P95, DOI 10.1016/j.biomaterials.2017.02.001, DOI	2017	5.68	2020	2021	
Machino R, 2019, ADV HEALTHC MATER, V8, P0, DOI 10.1002/adhm.201800983, DOI	2019	5.13	2020	2024	
Park JH, 2019, SCI REP-UK, V9, P0, DOI 10.1038/s41598-019-38565-z, DOI	2019	6.21	2021	2024	

Fig. 5 **a** The top 25 most prolific authors. **b** The visualization of co-cited authors. **c** co-authorship and **d** bibliographic coupling maps of authors. Representation of **e** highly co-cited references and **f** references with great centrality in the knowledge mapping of the reference co-citation analysis. **g** The top 25 references in citation bursts

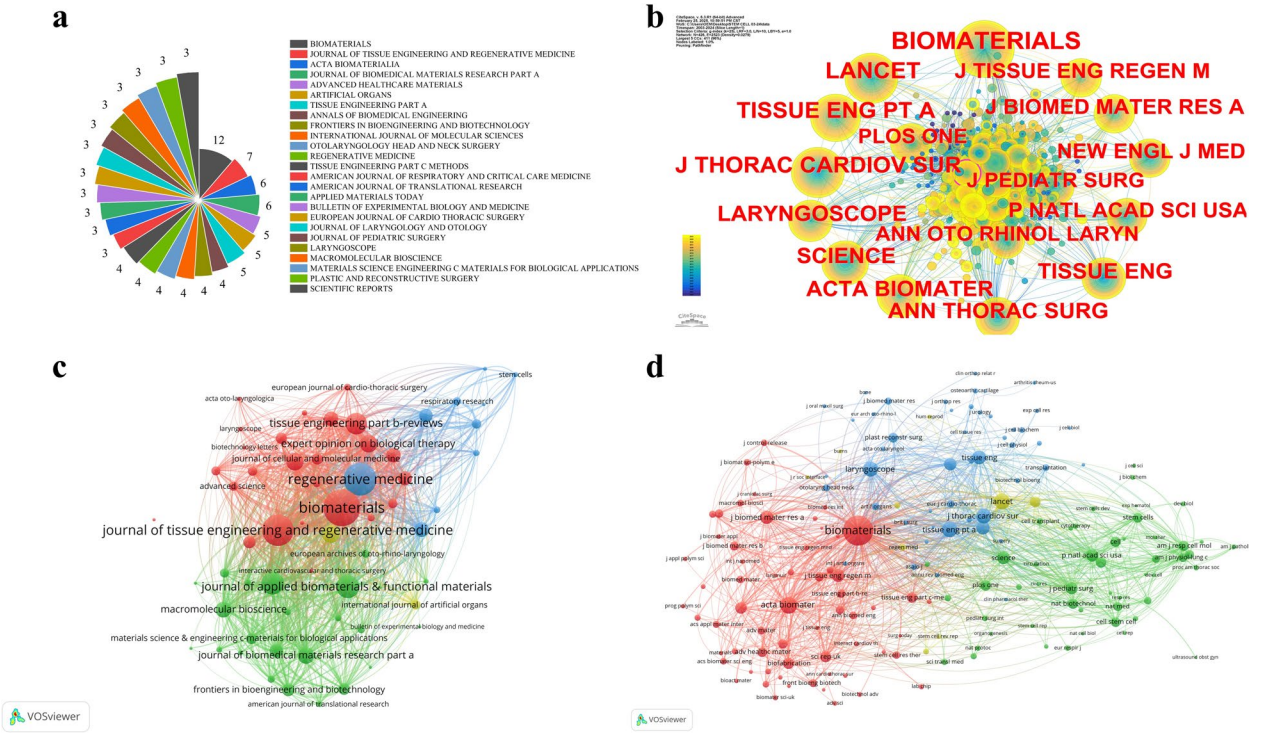


Fig. 6 High-contribution journal presentations on stem-cells research in the fabrication of a tissue engineering trachea. **a** The top 25 journals in terms of the number of relevant papers published. Journal co-citation knowledge network generated by **b** CiteSpace and **d** VOSviewer respectively. **c** Visual knowledge map of bibliographic coupling analysis of journals

similar to native trachea" [55] published in Biomaterials has been cited more than 120 times, and the citation burst intensity of this article was among the top 10 with the citation burst time lasting for four years (2017–2020, Fig. 5g). Above all, the latest works to be published on Biomaterials in the future deserve the attention of relevant scholars in this area.

Keyword analysis

Keywords can reflect the topics of the articles to a large extent. In this study, we conducted co-occurrence analysis, cluster analysis and burst analysis of keywords from publications related to stem cells in TTE research to explore the research hotspots and frontiers.

The keywords presented in Fig. 7a and b reflect the hot and focused research topics in the study of constructing tissue-engineered trachea grafts based on stem cells. The ten keywords with most counts of co-occurrence were, in order: "tissue engineering" (112 times), "stem cells" (110 times), "in vitro" (83 times), "mesenchymal stem cells" (78 times), "scaffolds" (68 times), "transplantation" (65 times), "regenerative medicine" (47 times), "replacement" (44 times), "regeneration" (44 times), and "reconstruction" (44 times), as well as the top 10 keywords by the centrality were as follows: "bone marrow" (0.24), "cartilage" (0.21), "in vitro" (0.18), "mesenchymal stem cells" (0.17), "differentiation" (0.14), "expression" (0.14), "stem cells" (0.13), "extracellular matrix" (0.13), "regeneration" (0.12), and "transplantation" (0.1) (same as "replacement" and "reconstruction"). It can be seen that there had been a strong focus on exploring suitable scaffolds, applying stem cells from different sources in the development of tissue-engineered tracheae, as well as emphasizing in vitro validation. The keywords "cartilage" and "differentiation" stood out with their high centrality ranking, indicating that research on tracheal cartilage regeneration and stem cell differentiation was an important step in the development of this research field.

Additionally, the keywords were mainly divided into twelve clusters through CiteSpace (Fig. 7c). And the clusters were structurally significant and credible, because the modularity Q was 0.5015 (>0.3) and the weighted mean silhouette S was 0.7724 (>0.7). They were "regeneration" (Cluster #0), "gene expression" (Cluster #1), "epithelial cells" (Cluster #2), "regenerative medicine" (Cluster #3), "3d printing" (Cluster #4), "fetal stem cells" (Cluster #5), "stenosis" (Cluster #6), "chondrogenic differentiation" (Cluster #7), "extracellular matrix" (Cluster #8), "clinical transplantation" (Cluster #9), "cartilage" (Cluster #10), "nanofibrous scaffolds" (Cluster #11), in order. As can be seen from these, scholars have been committed to exploring the construction of chondrified biomaterial scaffolds for the reconstruction of the trachea, based on the ability of stem cells to differentiate into chondrocytes

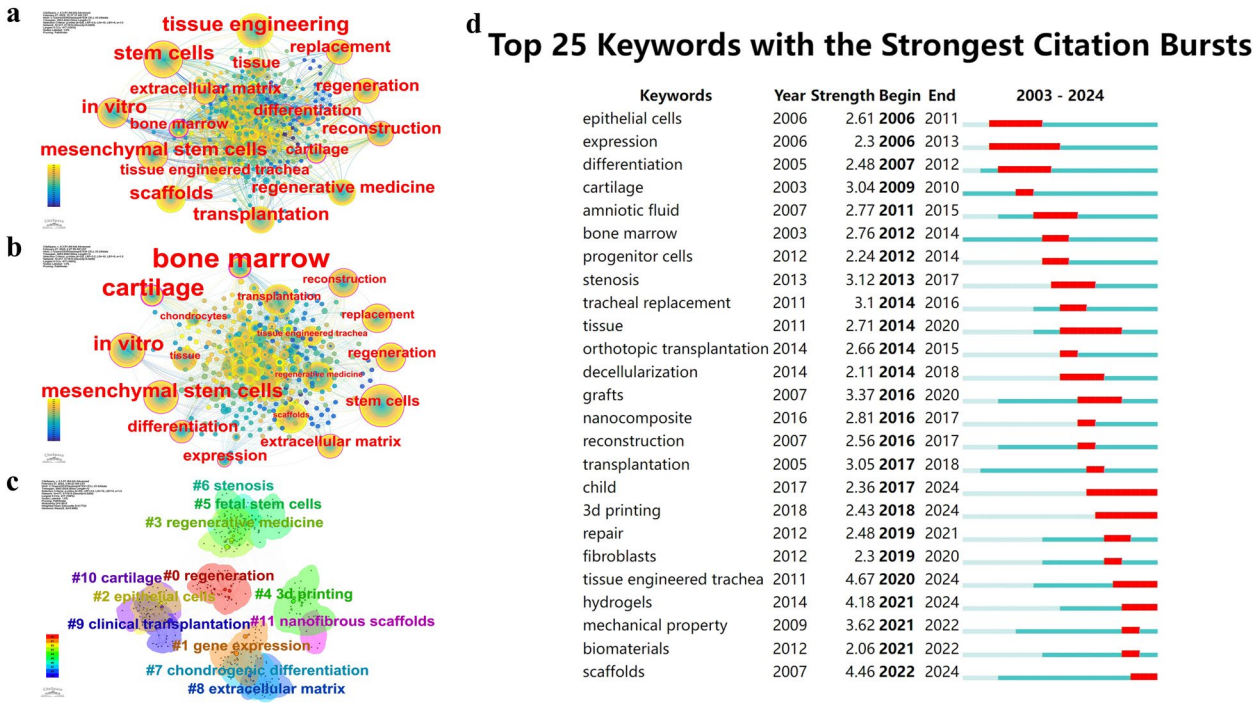


Fig. 7 Keyword analysis visualization. Display of **a** high-frequency co-occurrence keywords and **b** high-centrality keywords in the keyword co-occurrence map. **c** Visualization mapping of keywords clustering. **d** The top 25 keywords in citation bursts

and promote cartilage regeneration. This is because chondrification of tissue-engineered tracheal substitutes is one of the keys to successfully reconstructing the tracheal mechanical properties, which is an indispensable part of maintaining tracheal patency without collapse [56–58]. The studies [59–62] with citation bursts lasting until 2024 in the list of the top 25 references with the strongest citation bursts (Fig. 5g) focused on the research aspect of tracheal cartilage regeneration, which largely reveals the current research focus and the speculative future research trajectory. Therefore, it may be the reason that the keyword "mechanical property" was among the top 5 keywords with the strongest citation burst strength (Fig. 7d). In-depth studies are essential to be conducted, although stem cells have been shown to be effective in the regeneration of tracheal cartilage [59, 60, 63]. And this not only involves the exploration of cartilage differentiation strategies of stem cells, but also involves the selection of scaffold materials, the method of preparing scaffolds, and the strategy of combining stem cells with scaffolds. Furthermore, in order to achieve functional reconstruction of the trachea, the researchers also focused on the tracheal epithelial repair performance of the constructed tissue-engineered trachea, as well as the occurrence of stenosis.

The citation bursts for the keywords "3d printing", "tissue engineered trachea", "hydrogels", and "scaffolds" remained active until the year 2024, which suggested that using hydrogel as the raw material for scaffolds or as a carrier for stem cells to construct tissue-engineered trachea through 3D printing is a current hot spot and frontier in this field.

With the continuous in-depth exploration of stem cells, various sources of stem cells have been gradually developed for tissue regeneration, such as mesenchymal stem cells derived from adult tissues like bone marrow, fat, umbilical cord, and dental pulp; embryonic stem cells; iPSCs fabricated by gene editing [64–67]. In the above bibliometric and visual analysis, it is revealed that extensive research has been conducted on constructing tissue-engineered tracheal grafts for tracheal reconstruction using stem cells. And this includes not only the introduction of multiple types of stem cells for research, but also the exploration of stem cell-based strategies for constructing tissue engineered trachea grafts.

Stem cell types used in TTE

Stem cell types are diverse, and Fig. 8 and Table 1 shows the various stem cell types that have been studied in TTE.

Mesenchymal stem cells (MSCs)

MSCs, which are pluripotent stem cells capable of self-renewal and differentiation in multiple directions, can be extracted from a variety of tissues such as bone marrow, embryos, umbilical cords and adipose tissue [139–141]. Besides, the powerful immunoregulatory functions of MSCs have made them a popular choice for treating immune/inflammatory diseases and aiding in

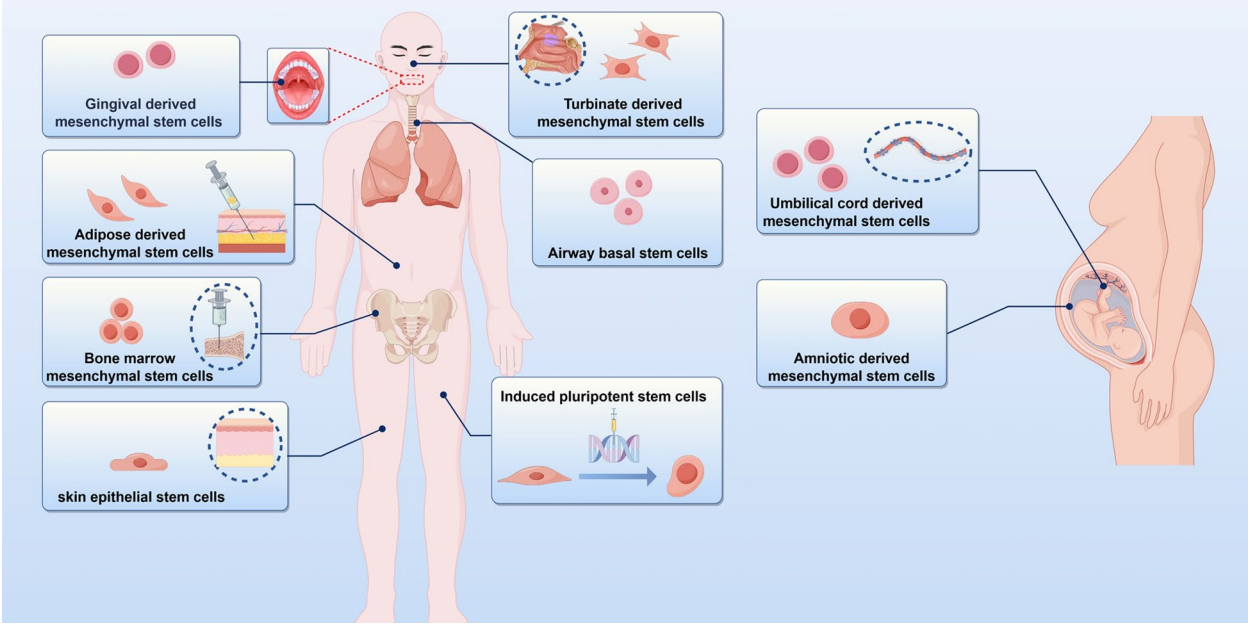


Fig. 8 The stem cell types that have been studied in TTE (Created by Figdraw)

tissue injury repair, yielding numerous positive effects [142–145]. And the keyword "mesenchymal stem cells" showed high co-occurrence frequencies and centrality both in keyword analysis, demonstrating that MSCs have aroused wide attention from researchers in TTE and have become one of the highly regarded sources of stem cells in TTE.

Bone marrow mesenchymal stem cells (BMSCs)

BMSCs were the first type of mesenchymal stem cells to be discovered, and they can be easily isolated from bone marrow through adherent or gradient centrifugation methods [68, 69, 146]. Researchers utilized BMSCs to assess the cytotoxicity and biocompatibility of tracheal scaffolds in vitro, demonstrating the potential for in vivo transplantation [147, 148]. Furthermore, multiple studies have provided evidence for the ability of BMSCs to differentiate into chondrocytes [81, 149–154]. This makes BMSCs potentially useful for chondrogenesis in tissue-engineered tracheal scaffolds [155–158]. And Yang Y et al. [80] successfully induced BMSCs to differentiate into chondrocytes in a circular hydrogel containing transforming growth factor (TGF)-β1 and kartogenin (KGN), resulting in the formation of a tracheal cartilage ring with good mechanical properties both in vitro and in vivo. Additionally, BMSCs can promote wound healing and tissue regeneration. Hydrogels containing BMSCs and chondrocytes showed better chondrogenic effects than those containing only chondrocytes [82]. And Ott LM et al. [74] found that grafts inoculated with BMSCs had better vascularization and epithelialization effects, as well as promoted chondroid structure formation. This may be related to the paracrine role of BMSCs, which have been shown to promote wound healing and cartilage regeneration through extracellular vesicles secreted by BMSCs [159–163]. Therefore, BMSCs have received much attention from researchers in the field of stem cells in TTE, and the high centrality shown by "bone marrow" in keyword analysis has also been revealed.

Adipose-derived mesenchymal stem cells (ADMSCs)

ADMSCs are easily and safely obtained. They can typically be isolated through collagenase digestion of adipose tissue derived from subcutaneous fat or adipose aspirate [85–87]. Hashemibeni B et al. [91] successfully induced the differentiation of ADMSCs into chondrocytes in an alginate gel containing TGF-β3 and achieved the regeneration of cartilage after implantation in tracheal cartilage defects. Additonally, ADMSCs co-cultured with chondrocytes showed a trend of chondrocyte differentiation, and chondroid structure formation occurred post-transplantation in vivo after the co-cultured mixed cells were inoculated into tracheal scaffolds [89]. Furthermore, the presence of ADMSCs can enhance the healing of the tracheal graft and the surrounding airway wall, expedite the vascularization process of the graft, decrease anastomotic stenosis, and increase the survival rate [43, 90]. This makes ADMSCs exhibit tremendous potential for applications in tracheal tissue engineering.

Umbilical cord derived mesenchymal stem cells (UCMSCs)

Umbilical cords are seen as a rich source of MSCs because of their ease of collection and lack of ethical concerns [93]. And UCMSCs are obtained from the umbilical

Table 1 Stem cell types used in TTE and stem cell based tracheal construction strategy for TTE

Types of stem cell		Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
MSCs	BMSCs	Bone marrow	1. The bone marrow used for BMSCs extraction is obtained by invasive puncture; 2. Discovered earliest and studied extensively; 3. Possess multi-directional differentiation potential, immune regulatory characteristics and self-renewal ability; 4. The proliferation ability is relatively weak and decreases with the increase of the donor's age. [68–72]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Ring-shaped type-II atelocollagen– Chondroitin sulfate scaffold (COL II–CS)	–	After 12 weeks of culture in chondrogenic medium of COL II–CS inoculated with BMSCs, chondro-specific extracellular matrix (ECM) deposits similar to the cartilage ring of the natural trachea were observed, which showed some degree of elasticity	By stacking chondrogenic COL II–CS rings with collagen I rings alternately, tracheal grafts with biomimetic mechanical properties were successfully generated in vivo, whose structure and function were similar to those of natural gas tubes	[73]
					Decellularized trachea	–	–	The presence of BMSCs promotes the formation of vascular networks around the acellular tracheal patch, promotes submucosal and mucosal regeneration, and exhibits anti-inflammatory activity. There was no significant stenosis 4 weeks after surgery	[31]
					electrospun poly(D, L-lactide-co-glycolide) (PLGA)(outer)–polycaprolactone (PCL)(inner) containing TGF-β3	–	–	The presence of BMSCs promoted the formation of graft cartilage and the regeneration of tracheal mucosa	[74]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				Decellularized trachea modified by collagen	–	After 4 weeks of culture in vitro, the expression of chondrogenic genes (COL2, aggrecan, Sox-9) of BMSCs inoculated on collagen-modified acellular trachea significantly increased	The results of subcutaneous embedding in nude mice showed that chondroid structures of the collagen-modified acellular trachea, seeded with BMSCs, formed after 8 weeks of embedding. The embedding time was extended to 16 weeks, and mature cartilage was observed. And the mechanical properties of the grafts improved, comparable to those of the original trachea	[75]
				Decellularized articular cartilage matrix	–	–	New cartilage formation, new blood vessel formation, and ciliary epithelium regeneration were noted after 10 weeks of in vivo transplantation	[76]
				Decellularized dermis matrix	–	–	The graft was incubated in the cartilage differentiation medium for one week before transplantation. After 3 months of in vivo transplantation, cartilaginous structure formation, neovascularization, and lumen epithelialization were observed in the graft	[77]
				Electrospun PCL-collagen scaffold containing TGF-β3	–	–	BMSCs exhibited cartilage differentiation potential on TGF-β3-loaded electrospinning scaffolds. After two months of orthotopic transplantation, it was observed that the graft had cartilage and its surface was covered by ciliated epithelium	[78]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				PLGA-poly(trimethylene carbonate) Membrane– Decellularized trachea	–	–	BMSCs were induced to differentiate into chondrocytes and then inoculated onto the membrane surface. Cartilage regeneration was observed 4 weeks after graft transplantation, and the mechanical properties were improved	[79]
			Encapsulating stem cells with hydrogel for constructing tissue engineering trachea	–	Sulfated hyaluronic acid–gelatin hydrogel containing TGF-β1 and kartogenin	The expression levels of chondrogenic proteins (COL II and SOX9) and key chondrogenic marker genes (SOX9, ACAN, and Col2a1) were significantly levate after 14 days of incubation in vitro	After in vivo transplantation, the cartilage ring with good mechanical properties was regenerated, the defect was fused with the natural gas tube, and the lumen was unobstructed	[80]
				3D printed PCL scaffold	Gelatin methacrylate (GelMA)	The deposition of cartilage components was observed after 21 days of culturing in chondrogenic medium in vitro	Four weeks after transplantation in vivo, small cartilaginous structures were observed	[81]
				3D printed PCL scaffold	Sodium alginate hydrogel	The chondrogenic genes (ACAN, Col1α1, Col2α1, and SOX9) of BMSCs cultured in chondrogenic medium were significantly elevated in a short period of time compared with BMSCs cultured without induced differentiation	New cartilage formation, epithelialization, and vascularization were observed in the grafts containing BMSCs (cultured in chondrogenic medium) 12 weeks after transplantation in vivo	[59]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				–	photocurable poly(ethylene glycol)(PEG)–PCL hydrogel	Compared to PEG–PCL hydrogels containing only chondrocytes, the mRNA levels of chondrogenic genes (SOX9, ACAN, Col2α1) in PEG–PCL hydrogels containing BMSCs (30%) and chondrocytes (70%) were higher, and more cartilage matrix components were produced	–	[82]
			Stem cell sheets reconstructing tracheal components	Decellularized trachea	–	In vitro induction of BMSCs sheet into endothelial differentiation for 21 days, a significant increase in angiopoietin-1 was observed	Microvascular formation was enhanced by grafts covered with BMSCs sheets	[83]
			Stem cell spheroid reconstructing tracheal components	–	–	Strong chondrogenic ability was observed in chondrocyte spheroids containing BMSCs	35 days after transplantation, deposition of cartilage matrix components was observed	[61]
				–	–	The scaffold-free tracheal grafts constructed from spheroids containing BMSCs, chondrocytes, and endothelial cells showed certain mechanical properties after 28 days of culture in vitro	After 23 days of transplantation, the formation of cartilage and blood vessels was observed	[60]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				–	–	BMSCs spheroids differentiated into chondrospheroids in vitro by culture medium containing TGF-β3 and fibroblast growth factor-2. Through 3D bioprinting technology, the bionic tracheal graft composed of a horseshoe-shaped cartilage tube (generated by BMSCs-derived cartilage spheroids) and a smooth muscle membrane (generated by smooth muscle cell spheroids) was successfully constructed after 3 weeks of culture in vitro	After 4 weeks of in vivo transplantation, the graft can still maintain its shape and stiffness. The expression of COL II was strongly positive in the cartilage area	[84]
ADMSCs	Adipose	1. ADMSCs are easy to obtain and minimally invasive to the donor because they can be extracted from the adipose tissue obtained through liposuction; 2. Possess multi-directional differentiation potential, self-renewal ability and good paracrine function; 3. Its immunomodulatory ability is superior to that of BMSCs, and the early proliferation ability is relatively strong [85–88]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Electrospun poly (L-lactic acid) (PLLA) scaffold	–	–	The collagen fibers regenerated after in vivo transplantation on a PLLA patch inoculated with ADMSCs were more orderly and showed a healthy histological tracheal structure	[43]
				Electrospun PCL- collagen membrane– Decellularized aorta–	–	The expression of COL II increased after 21 days of in vitro culture	After 4 weeks of muscle embedding, the deposition of cartilage components was observed	[89]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				Decellularized trachea	–	–	The graft healed well with the surrounding tissue and the ciliary epithelium regenerated	[90]
			Encapsulating stem cells with hydrogel for constructing tissue engineering trachea	Collagen sponge containing polypropylene mesh	Neutral atelocollagen	–	Grafts containing ADMSCs exhibited better morphological and functional regeneration of the tracheal epithelium	[30]
				–	Alginate hydrogel containing TGF-β3	After 2 weeks of culture in chondrogenic medium, it was observed that ADMSCs in alginate hydrogel differentiated into chondrocytes and deposited cartilage matrix	Cartilage formation in the tracheal cartilage defect area was observed 8 weeks after transplantation	[91]
			Stem cells derived exosomes promoting repair of trachea construction	Decellularized trachea	GelMA	ADMSCs -derived exosomes (ADMSC-EVs) enhanced the proliferation and migration of human bronchial epithelial cells (HBE)	Three months after in vivo transplantation, the grafts were not stenosed and the regeneration of pseudostratified ciliated columnar epithelium was enhanced	[92]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
UCMSCs	Umbilical cord	1. The acquisition of UCMSCs is non-invasive to donors and lacks ethical risks; 2. Possessing multi-directional differentiation potential, as well as extremely strong and stable proliferation ability; 3. Possess outstanding paracrine function and can secrete a variety of cytokines; 4. Lack of limitation of donor age; 5. Have low immunogenicity and strong function of promoting tissue repair–regeneration, suitable for large-scale preparation and have broad prospects for clinical application [93–96]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Decellularized amniotic membrane (DAM)	–	–	After chondrogenic differentiation culture, UCMSCs were seeded on DAM and successfully promoted cartilage regeneration	[32]
				Electrospun poly (lactate-caprolactone) -collagen nanofiber scaffold containing TGF-β3	–	After 3 weeks of in vitro culture of UCMSCs on scaffolds, increased expression of the chondrogenic genes (Sox-9 and COL2) and deposition of cartilage matrix components were observed	–	[97]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
AMSCs	Amniotic membrane	1. AMSCs have no ethical conflict because they are extracted from discarded amniotic tissue after delivery; 2. Lack of limitation of donor age; 3. Possessing multi-directional differentiation potential, low immunogenicity, and outstanding paracrine performance; 3. Compared with MSCs derived from adult tissues, AMSCs have a stronger proliferation ability; 4. Lack of issues regarding the donor's age [98–103]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Decellularized trachea	–	–	The decellularized tracheae seeded with AMSCs were incubated in chondrogenic medium for 7 days, and the tracheal defects were repaired. The results showed that the grafts had deposited cartilage matrix components, and the cartilage lacunae of the acellular trachea were filled with cells. In addition, the presence of AMSCs effectively promoted the regeneration of the tracheal epithelium	[104]
GMSCs	Gingiva	1. GMSCs is easy to obtain and causes little trauma to the donor; 2. Possessing multi-directional differentiation potential and paracrine performance; 3. Have a strong immune regulation and anti-inflammatory effects [105–110]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Decellularized trachea	–	–	The infiltration of immune cells was inhibited and the vascularization of the graft was accelerated	[111]
			Stem cells derived exosomes promoting repair of trachea	Decellularized cartilage extracellular matrix (DCM)	–	The proliferation and migration of HBE were promoted with the presence of GMSCs-derived exosomes (GMSC-EVs)	DCM containing GMSC-EVs promoted the proliferation of tracheal progenitors—basal epithelial cells and promoted the rapid regeneration of ciliated columnar epithelium and blood vessels at the site of a tracheal defect	[34]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
TMSCs	Turbinate	1. The surgically removed turbinate tissue can be used to extract TMSCs; 2. Have multi-lineage differentiation, paracrine function, and immunomodulatory properties 3. The donor's age and allergic condition have little effect on the characteristics of TMSCs [112–116]	Stem cell sheets reconstructing tracheal components	PCL– Collagen scaffold	–	–	After 28 days of in vivo transplantation, the luminal surface was observed to be covered by epithelial cells	[117]
				3D printed PCL scaffold	Collagen gels	After tracheal epithelial induction culture in vitro, the TMSCs sheet showed potential to differentiate into epithelial cells	After 4 weeks of in vivo transplantation, it was observed that the luminal surface of the graft was covered by mature and highly ciliated epithelium	[118]
				3D printed PCL scaffold	decellularized tracheal mucosa matrix hydrogel	–	After 2 months of in vivo transplantation, complete re-epithelialization was achieved on the entire lumen surface, but mild stenosis due to granulation tissue proliferation was observed	[119]
			Encapsulating stem cells with hydrogel for constructing tissue engineering trachea	3D printed PCL scaffold	HyStem-HP hydrogels containing TGF-β1	After two weeks of culture in vitro, cartilage formation and smooth muscle formation were observed	–	[120]
ABSCs	Airway epithelium	1. Self-renewal ability is strong; 2. ABSCs are involved in the repair of airway epithelium and can differentiate into various cell types of airway epithelium, including ciliated cells, goblet cells, brush cells, etc [121–123]	Encapsulating stem cells with hydrogel for constructing tissue engineering trachea	Decellularized trachea	Type I collagen hydrogel	–	One week after transplantation, signs of revascularization were observed and keratin-positive cells were found throughout the scaffold	[124]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
SESCs	Epidermis of skin	1. SESCs has the ability of self-renewal and the potential of differentiation; 2. It plays an important role in the repair of skin injury [125–127]	Stem cell direct-seeded scaffolds for tracheal reconstruction	Polypropylene mesh–dermal matrix scaffold	–	–	Ciliated epithelium regeneration and chondrogenesis of the trachea were observed after 5 months of transplantation, and SESCs were demonstrated to differentiate into tracheal epithelium and chondrocytes	[128]
iPSCs	Somatic cells treated with genetic reprogramming	1. iPSCs possess pluripotency similar to that of embryonic stem cells and can differentiate into all cell types derived from the three germ layers (endoderm, mesoderm, and ectoderm); 2. Have an extremely strong self-renewal ability and can multiply infinitely outside the body; 3. iPSCs can be induced from the recipient's own somatic cells (such as skin fibroblasts), which can avoid immune rejection and achieve personalized treatment; 4. There is a risk of tumor formation [129–134]	Stem cell direct-seeded scaffolds for tracheal reconstruction	silk fibroin–collagen membrane	–	iPSCs were successfully induced to differentiate into Airway progenitor cells in vitro and showed the ability to differentiate into ciliated epithelium	The graft was covered by a pseudostratified ciliated columnar epithelium	[33]
				Decellularized trachea	–	iPSCs were successfully induced and differentiated into ABSCs in vitro, and possessed the molecular and functional phenotypes of ABSCs	After 3 weeks of in vivo transplantation of acellular trachea containing iPSC-derived ABSCs (iPSC-ABSCs), it was observed that the iPSC-ABSCs differentiated into tracheal epithelium and pseudostratified ciliated columnar epithelium formed on the graft surface	[41]

Table 1 (continued)

Types of stem cell	Source	Characteristics	construction strategy	Scaffold	Hydrogel material	In vitro effect	In vivo effect	References
				Decellularized trachea	–	–	Regeneration of with a large number of ciliated cells was observed, but the presence of undifferentiated iPSCs proliferated to form cell masses causing airway stenosis	[135]
				electrospun PCL nanofibers (inner)–3D-printed PCL microfibers (outer)	–	iPSCs were successfully induced to differentiate into MSCs and chondrocytes	New cartilage and ciliated columnar epithelium were effectively formed at the defect site, and no granulation was found to grow into the tracheal lumen	[136]
			Encapsulating stem cells with hydrogel for constructing tissue engineering trachea	Collagen sponge containing polypropylene mesh	Type I collagen gels	The embryonic bodies (EB) formed by iPSCs were differentiated into tracheal epithelial cells by differentiation medium and air–liquid interface culture, and ciliary movement was observed	The regenerated epithelial structure from the grafts of the trachea containing iPSCs-derived tracheal epithelial cells was similar to that of normal epithelium	[137]
				Collagen sponge containing polypropylene mesh	Type I collagen gels	After incubating the grafts in chondrogenic medium for 6 weeks, the iPSCs in the grafts differentiated into chondrocytes and the grafts showed chondrogenesis	The results of in vivo transplantation showed that the grafts incubated with in vitro chondrogenic medium could form cartilage tissue in vivo, but tumor formation was also observed in some animals	[138]

cords of fetal appendages, with no donor age limitations, and show no decrease in proliferation and differentiation capacity or weakened immunoregulatory function due to donor aging [94, 95]. Additionally, UCMSCs, compared to BMSCs and ADMSCs, showed better adhesion and proliferation efficiency [164, 165]. The ability of UCMSCs to differentiate into chondrocytes in vitro has been confirmed [166, 167]. And, they successfully differentiated into chondrocytes in vitro on a tracheal scaffold coated with TGF-β3 [97]. Furthermore, the tracheal window defect repair experiment showed that a scaffold recellularized with chondrocytes, which were differentiated from UCMSCs, formed cartilaginous structures closer to those of a normal trachea [32]. Thus, utilizing UCMSCs as seed cells may be a promising approach for constructing tissue-engineered tracheal cartilage.

Amniotic derived mesenchymal stem cells (AMSCs)

AMSCs are derived from the amniotic membrane, one of the fetal appendages, without the need for the donor to

undergo invasive injury, as is the case with adult MSCs [98]. And it has been reported that AMSCs, along with the exosomes they release, have shown considerable potential for tissue repair and regeneration [99, 168–170]. The reconstruction of the epithelium was found to be better when scaffolds were seeded with AMSCs, as opposed to scaffold implantation alone [104]. Moreover, AMSCs were discovered to exhibit chondrogenic properties comparable to those of UCMSCs [100]. Consequently, it is unsurprising that scaffolds seeded with AMSCs in this study displayed the deposition of cartilage matrix [104].

Gingival derived mesenchymal stem cells (GMSCs)

Gingiva, a unique keratinized mucosal tissue, has the characteristic of rapid wound healing and no scarring [171, 172]. And the stored GMSCs within it are thought to be an important reason for maintaining this characteristic [173]. By examining the orthotopic transplantation experiment, it was evident that GMSCs were instrumental in expediting the vascularization of decellularized tracheal scaffolds [111]. Among them, the paracrine function of GMSCs was a crucial aspect that cannot be disregarded. The research conducted by Zeng T et al. [34] demonstrated that the inclusion of exosomes derived from GMSCs accelerated airway epithelial regeneration and angiogenesis of the graft, effectively reducing scar stenosis, and ultimately resulting in a notable enhancement in prognosis. What's more, GMSCs can be obtained from the gingiva using minimally invasive surgery, which is both safe and minimally invasive [173]. And They are advantageous as seed cells.

Turbinate derived mesenchymal stem cells (TMSCs)

TMSCs may be another potential source of MSCs, as the MSC characteristics are not absent in TMSCs, such as multi-lineage differentiation, paracrine function, and immunomodulatory properties [112]. And the age of the donor, the size of the donor's turbinate, and the donor's allergic condition had little effect on the characteristics of TMSCs [113–115]. By combining TMSCs with tracheal scaffolds, researchers not only effectively inhibited graft immune rejection, but also promoted graft vascular and organ epithelial regeneration [117–119]. However, it is important to note that further preclinical trials are necessary before the clinical implementation of TMSCs.

Airway basal stem cells (ABSCs)

ABSCs, considered as airway-resident stem cells, are an important component of the airway epithelium [121, 174]. They can self-renew and differentiate into ciliated cells and secretory cells to repair the airway after acute injury, playing a crucial role in maintaining the airway epithelium [122, 123]. In addition to being isolated from epithelial tissue harvested by bronchoscopic forceps, ABSCs can also be harvested from tracheal aspirates to reduce tissue damage, as described by Amonkar GM [175]. Besides, researchers discovered that ABSCs were able to expand and maintain their stemness in vitro by binding them to ROCK inhibitors and 3T3 fibroblasts [176, 177]. On this basis, Hamilton NJI et al. [124] successfully inoculated ABSCs into collagen scaffolds and prepared tissue-engineered tracheal scaffolds for epithelial regeneration, preserving their capacity to grow and differentiate. Excitingly, they observed keratin-positive epithelial cells just 7 days after in-situ transplantation, indicating the potential of ABSCs for epithelial construction. Furthermore, ABSCs can significantly inhibit the proliferation of granulation tissue, reducing stenosis of the tracheal lumen, which is crucial for the prognosis of recipients undergoing tracheal reconstruction [178].

skin epithelial stem cells (SESCs)

The skin, being the largest organ in the human body, exhibits a robust capacity for self-repair following injury [179]. And SESCs have been identified as a key element of the skin epithelium, contributing significantly to the maintenance of skin integrity [180]. This is a result of SESC's attributes, such as self-updating and differentiation [181]. Moreover, Kim JH et al. found in vivo experiments that SESCs can retain self-renewal and differentiation abilities after transplantation, and can effectively differentiate into tracheal epithelial cells and chondrocytes [128].

Induced pluripotent stem cells (iPSCs)

In 2006, Yamanaka created iPSCs by introducing four transcription factors—Oct3/4, Sox2, c-Myc, and Klf4—into fibroblasts [129]. And iPSCs, being pluripotent stem cells, have the ability to differentiate into cells of three germ layers, and their pluripotent capabilities are comparable to those of embryonic stem cells [130, 131]. Zhou Q et al. [135] observed that iPSCs were able to re-epithelialize tracheal scaffolds through their differentiation potential. However, they noted that in one recipient, the iPSCs did not differentiate as expected and instead grew into clusters, ultimately causing airway stenosis. For the directional differentiation of iPSCs into tracheal component cells to construct a trachea stably and successfully, Hawkins FJ et al. [41, 182] attempted to prompt iPSCs to differentiate into tissue-specific ABSCs. And the resulting cells successfully exhibited the molecular and functional characteristics of ABSCs, which have potential in constructing the re-epithelialization process of tissue-engineered trachea. Moreover, certain researchers have successfully prompted iPSCs to differentiate into airway epithelial cells or chondrocytes for reconstructing tracheal epithelium or cartilage, and have proven the

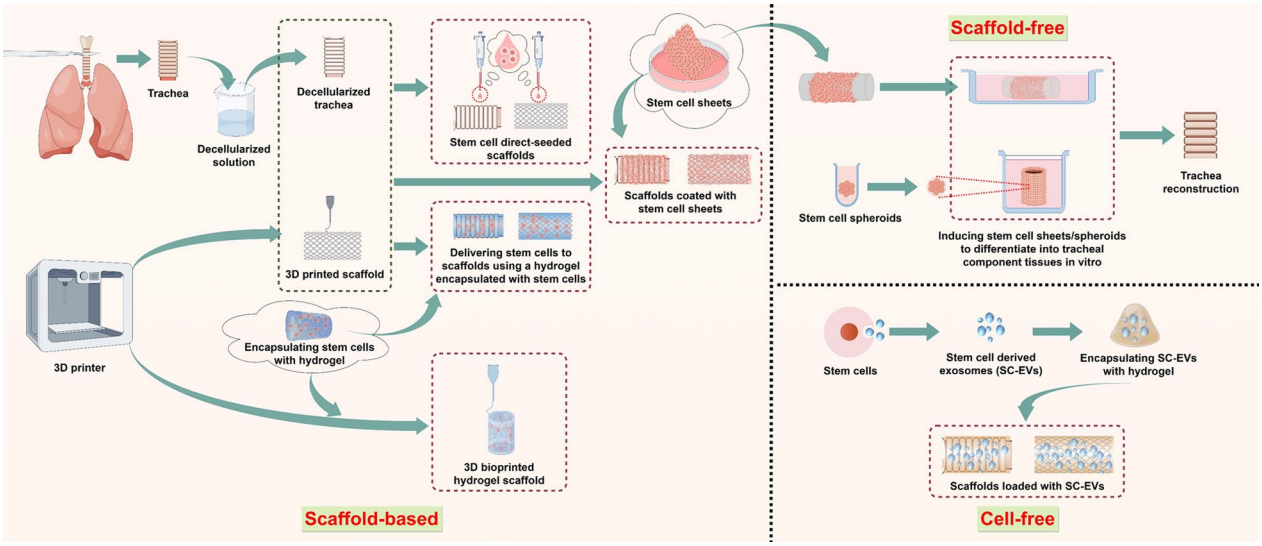


Fig. 9 Schematic diagram of the design and manufacture of tissue-engineered trachea based on stem cells (Created by Figdraw)

efficacy of iPSC-derived airway epithelium and iPSC-derived chondrocytes in tracheal repair [33, 137, 138]. Despite the considerable economic and time investments required for iPSC preparation, iPSCs have the potential to be an ideal resource for tracheal tissue engineering [183]. However, it is crucial to address the tumorigenic risks associated with iPSCs before considering their application in clinical settings. Furthermore, in this review, we summarize the types of stem cells that have been applied in TTE studies and the construction strategies for stem cell-based tissue-engineered trachea.

Stem cell based tracheal construction strategy for TTE

Figure 9 and Table 1 illustrate several methods for designing and fabricating tissue-engineered tracheae based on stem cells.

Stem cell direct-seeded scaffolds for tracheal reconstruction

The trachea is a tubular organ with specific mechanical characteristics. In cases of tracheal defects, scaffold repair is essential for maintaining effective ventilation [1]. Hence, utilizing a tracheal scaffold to direct-deliver stem cells for trachea reconstruction is a popular approach among TTE researchers.

The extracellular matrix (ECM) is a vital component of tissues and organs, contributing significantly to tissue structure, microenvironment, and cell interactions. Attempts to decellularize the tissue to remove the cellular components and retain the effective ECM components as a biological scaffold for stem cell inoculation have shown great advantages [184]. Allogeneic trachea is one of the main sources used in the preparation of decellularized matrix for repairing tracheal defects. And studies have indicated that the acellular trachea showed low immunogenicity and good biocompatibility, allowing stem cells to adhere to the surface of the acellular trachea for proliferation and differentiation [185–187]. Direct seeding of stem cells on the surface of the acellular trachea significantly facilitated revascularization, epithelialization, and healing of the acellular trachea with surrounding tissues after transplantation [31, 90, 104]. Aiming to enhance BMSC adhesion efficiency, Xu et al. [75] employed collagen to modify the surface of decellularized trachea, which improved the adhesion efficiency of BMSCs and promoted the cartilage differentiation of BMSCs. Additionally, there are also other tissues used for MSC loading after decellularization. Shin et al. [76] designed a scaffold composed of decellularized articular cartilage matrix for MSC delivery and successfully reconstructed tracheal cartilage and ciliated epithelium 10 weeks after repairing tracheal window defects in rabbits. Al-Ayoubi et al. [77] inoculated BMSCs on acellular dermis to construct tissue-engineered tracheal patches. The patches were then incubated in the cartilage differentiation medium for one week and used to repair tracheal window defects. After 3 months of in vivo transplantation, cartilaginous structure formation, neovascularization, and lumen epithelialization were observed in the graft.

In order to customize tracheal scaffolds, researchers have been attempting to use artificial or natural materials through electrospinning, 3D printing, and other technologies to prepare bionic tracheal scaffolds, which can then be combined with stem cells to build tissue engineering tracheal grafts [78, 188, 189]. And the composition, mechanical properties, porosity and biocompatibility of the scaffolds prepared through it are related to the adhesion, proliferation and differentiation of stem cells. Pan et al. [190] investigated the influence of pore

size of a 3D printed polycaprolactone (PCL) mesh scaffold on the proliferation of BMSCs. They also mentioned that increasing the surface hydrophilicity of the scaffold through surface modification contributes to the adhesion of BMSCs. This is significant for inoculating and delivering stem cells. In addition, tracheal scaffolds with suitable mechanical properties and good cell adhesion properties were prepared by mixing different biomaterials for the successful inoculation, proliferation and delivery of MSCs [79, 191]. 3D printed scaffolds and electrospun scaffolds make up for the deficiencies of each other, and the strategy of using them to construct hybrid scaffolds for loading stem cells has also been reported [136]. Using the ability of stem cells to differentiate into cartilage to cartilaginize scaffolds to achieve the mechanical properties required for transplantation is a promising strategy. For example, Xu et al. [73] seeded BMSCs on a ring-shaped scaffold composed of type-II atelocollagen and chondroitin sulfate. BMSCs on the ring-shaped scaffold differentiated into cartilage, and the scaffold was successfully cartilaginous after 12 weeks of in vitro culture in a chondrogenic medium. And the biomimetic trachea containing the cartilage rings had good mechanical properties and was well integrated with the surrounding tissues after transplantation.

Encapsulating stem cells with hydrogel for constructing tissue engineering trachea

Hydrogels are a type of hydrophilic gel with a three-dimensional network structure, which can be used to encapsulate high-density stem cells [192]. Natural or synthetic biodegradable materials are commonly used materials for preparing hydrogels [193]. The hydrogel composed of these materials has good biocompatibility and biodegradability, which can provide a suitable growth and differentiation environment for stem cells. And it can also gradually degrade during tissue repair to avoid long-term immune rejection reactions. Various strategies have been employed to develop synthetic biomimetic hydrogels with great swelling properties and similar structures to tissues [194]. Additionally, by adjusting the pH, temperature, degree of crosslinking, and composition ratio, the biophysical and biochemical properties of these hydrogels can be regulated to promote stem cell proliferation and differentiation [195, 196]. Hence, hydrogels have been significantly concerned in the field of stem cells in TTE in recent years, as evidenced by Fig. 7d. Mechanical properties of trachea substitutes for transplantation are an indispensable aspect. Naito H et al. [197] loaded MSCs with collagen-based hydrogels and successfully induced bone formation in vitro, effectively improving the mechanical properties of the hydrogel-based tracheal scaffold. However, tracheal cartilage regeneration is the ultimate goal of recovering tracheal mechanical properties and even reconstructing the trachea. Hydrogels have been shown to be excellent carriers for drug or cytokine loading and release [198]. Thus, some bioactive substances can be added to the hydrogel to promote the proliferation, migration, and differentiation of MSCs encapsulated in the hydrogel [199, 200]. For example, employing hydrogels containing bioactive substances (such as TGF-β1 and KGN) that enhance chondrogenic differentiation of stem cells for encapsulating stem cells to achieve tracheal chondrogenesis. Yang et al. [80] adopted this strategy and found that the hydrogel ring formed by pouring a composite hydrogel containing TGF-β1, KGN, and BMSCs achieved chondrogenesis and formed a cartilage ring. They eventually connected the cartilage rings in series to construct a bionic tracheal graft, which has mechanical properties comparable to the original trachea.

In recent years, the rise of 3D bioprinting technology has brought new hope for the construction of tissue engineering bionic tracheas based on hydrogel. Modeling the native trachea using a Computed Tomography scan, Ke et al. [120] divided it into the C-shaped cartilage region and the membrane region. And then, they prepared a cartilage bioink and a smooth muscle bioink by adding cartilage differentiation-inducing substances and smooth muscle differentiation-inducing growth factors to the hydrogel loaded with MSC, respectively. And the bionic trachea was constructed by filling the C-type cartilage area and the membrane area with 3D bioprinting, and the cartilage tissue and smooth muscle tissue were formed in the corresponding areas after culturing in vitro. Therefore, bioinks composed of stem cells and hydrogels have great potential for customizing suitable tissue-engineered tracheal grafts through 3D bioprinting [201, 202].

In conclusion, encapsulating stem cellswith hydrogels for cellularizing tracheal scaffolds is feasible and more effective than direct stem cell inoculation [203]. And the stem cell-loaded hydrogel can be delivered to different areas of the tracheal scaffold for the reconstruction of each tracheal component. For example, the inner and outer walls of the tracheal scaffold can be covered with stem cell-loaded hydrogels for epithelial regeneration and cartilage regeneration, respectively. In addition, stem cells from different sources can be delivered to the scaffold for repairing various components of the trachea using hydrogels. Hence, this strategy offers significant advantages for tissue engineering trachea construction.

Stem cell sheets/stem cell spheroids reconstructing tracheal components

With the proliferation and migration of adherent stem cells, they can fuse with each other in a culture dish. To isolate high-density cells while retaining intercellular junctions, ECM, and key proteins, researchers have tried

to use cell sheet technology to isolate stem cell sheets, which can help deliver large numbers of stem cells to the target area [204–206]. And it has been confirmed that stem cell tablets can differentiate into target cells by induction in vitro [207–209]. Yao et al. [83] found that BMSC sheets treated with a vasogenic induction medium showed a strong angiogenic ability, and decellularized tracheal scaffolds coated with the sheets showed a strong potential for angiogenesis. The tracheal epithelium grows on the surface of the endotracheal lumen and plays a role in resisting pathogenic invasion and clearing foreign bodies. Park et al. [118, 119] coated the TMSCs sheet on the inner surface of the tracheal stent, and the lumen surface was covered by mature and highly ciliated epithelium after orthotopic transplantation, effectively accelerating the regeneration of tracheal epithelium. Furthermore, leveraging the differentiation potential of stem cell sheets for developing scaffold-free tracheal transplantation alternatives may be a potential strategy in the reconstruction of tissue-engineered tracheae. Nonetheless, further research is warranted due to the distinctive nature of tracheal structure and function.

When stem cells are cultured in three dimensions, they tend to cluster and self-assemble to form multicellular spheroids [210]. Studies have shown that stem cell spheroids have many advantages, such as better intercellular and cell-extracellular matrix interactions, maintaining intrinsic phenotypic characteristics, promoting the expression of stemness markers, and secreting higher levels of cytokines compared to adherent stem cells [211]. Therefore, some researchers began to try to construct scaffold-free bionic tracheal grafts using spheroids containing stem cells and 3D bioprinting technology, which demonstrated the ability of chondrogenesis and vascularization in vitro and in vivo [60, 61, 84].

Stem cells derived exosomes promoting repair of trachea construction

Stem cell-derived exosomes are the main substances in the paracrine effect of stem cells and are currently one of the most widely studied and used exosomes [212]. They contain bioactive substances such as growth factors, mRNA, and microRNA, and can aid in tissue repair and immune regulation, which share similarities with MSCs therapeutics [213–215]. For example, they can affect the activation of macrophages to regulate the ratio of M1 polarization to M2 polarization to control the inflammatory response at the site of tissue injury, as well as to promote angiogenesis and cell migration [216, 217]. Importantly, stem cell-derived exosomes do not pose a possible risk of tumor formation due to their inability to self-replicate [218, 219]. This points to a promising approach for developing bioactive tracheal scaffolds that are cell-free, allowing for rapid repair of tracheal defects and restoration of tracheal function. Hydrogels are often used to load exosomes to prevent premature exosomal clearance and provide a sustained therapeutic effect [220]. Lu Y et al. [92] utilized Gelatin Methacryloyl-loaded ADMSC-derived exosomes (ADMSC-EVs) to coat the decellularized trachea surface. And their experiments in vitro and in vivo revealed that the presence of exosomes expedited the re-epithelialization of grafts when compared to the group without exosomes. Besides, it has also been reported that ADMSC-EVs can promote the repair of tracheal cartilage. Furthermore, some researchers developed a novel strategy for loading stem cell exosomes [34]. They co-cultured GMSCs with chondrocytes and assumed that GMSCs would release a rich amount of GMSC-derived exosomes (GMSC-EVs) in the ECM generated by chondrocytes. And the experimental results confirmed their hypothesis. Then, subsequent to a mild and concise decellularization treatment, GMSC-EVs were efficiently retained within the decellularized matrix, leading to the preparation of a decellularized matrix rich in GMSC-EVs.

Attempts at clinical transformation

In the early research stage, researchers mainly focused on exploring the feasibility of using stem cells combined with tracheal scaffolds for tracheal reconstruction. During this period, stem cells from various sources were included in the research, but BMSCs were mainly used as seed cells, and the initial exploration of scaffold materials was carried out. It was also during this stage that the strategy of combining stem cells with decellularized trachea demonstrated excellent tracheal reconstruction effects in animal models, which aroused the researchers' idea of conducting clinical trials. And Elliott MJ et al. [53, 221] successively reported two cases where decellularized trachea directly inoculated with BMSCs was applied to treat patients with severe tracheal stenosis. In the early postoperative follow-up, the phenomenon of graft collapse due to changes in mechanical properties was found in all cases, and no rapid epithelialization or tracheal cartilage regeneration occurred. This shows a huge gap compared with the therapeutic effect in the animal model. With the development and maturation of 3D printing technology, researchers have gradually begun to explore the use of 3D printing technology to construct personalized bionic tissue-engineered tracheal grafts that can be used to achieve tissue-specific structures. And this has made the interaction between stem cells and scaffolds more emphasized [222]. At present, with the continuous deepening of research and multidisciplinary cooperation (such as biomedicine, materials science, immunology, and computer science), some studies have successfully achieved the transplantation and functional recovery of tissue-engineered tracheae in animal models

such as rabbits and pigs [33, 223, 224]. However, ensuring the long-term stability and effectiveness of the graft remains a difficult point. And the previous attempts at clinical transformation have shown that the reconstruction of human trachea is more complex. Therefore, at present, the research on stem cell-based TTE is mainly at the stage of verifying the safety, efficacy and long-term function of related technologies through animal models and gradually transitioning to a preclinical phase.

Challenges and expectation of stem cells in TTE

Stem cells have been shown to have considerable potential in tracheal reconstruction, as these cells can not only differentiate into lost tracheal cells, but also promote tracheal repair by paracrine means. However, there are still great challenges for stem cell-based tissue-engineered tracheal grafts to enter the clinical translation stage.

Heterogeneity and tumorigenic risk of stem cells

The significant heterogeneity of stem cells cannot be ignored. The age and health status of the donor will affect the content, proliferation activity, and differentiation performance of stem cells [225]. For example, the density and cell function of MSCs derived from adult tissues such as bone marrow and fat will decrease with the aging of the donor [226, 227]. In addition, different isolation methods, culture conditions, and phenotypic loss during expansion can also lead to the heterogeneity of stem cells [228, 229]. These are some of the important reasons for the uneven and difficult reproducibility of the therapeutic effect of stem cells in preclinical and clinical studies. Therefore, in the process of transferring stem cells from the lab to clinical applications, it is necessary to continuously optimize the extraction and purification plans for stem cells (including accurately sorting out cell subpopulations with high differentiation potential and stability through flow cytometry technology, formulate and improve standardized protocols, etc.). And it is crucial to establish rigorous quality control standards such as stem cell purity standards, stem cell activity standards, and biosafety standards. Besides,checking if allogeneic stem cells have triggered receptor immune responses is necessary because immunogenicity exists even in MSCs with highly immune exempt.

Tracheal tissue repair requires a variety of cell types, including tracheal epithelial cells, smooth muscle cells, chondrocytes, etc. The self-renewal ability and multidirectional differentiation potential of stem cells are the basis of the therapeutic potential of stem cells in trachea reconstruction. However, it is precisely because of these properties that stem cells are at risk of forming tumors (such as teratoma) during transplantation in vivo. For example, if the transplanted stem cells (especially embryonic stem cells or iPSCs) are not fully differentiated, they may continue to proliferate in the body and form tumors [230]. Besides, genetic mutations in stem cells (e.g. those caused by long-term in vitro expansion or gene editing during iPSCs' preparation) or activation of abnormal proliferation signaling pathways in stem cells (e.g. those caused by the microenvironment at the transplant site or induced by excessive growth factors during differentiation) can increase the risk of tumor formation in stem cells [231–234]. Hence, to enhance the safety of stem cells and decrease the risk of tumorigenesis, rigorous preclinical safety assessments must be conducted, and the dynamics of stem cells post-transplantation should be observed over an extended period. Firstly, there exist differences in the differentiation potential and biological characteristics of stem cells from different sources [101, 235, 236]. In future studies, the characteristics of stem cells from different sources should be further refined to select the most appropriate seed cells for tracheal reconstruction. Secondly, gene editing technology can be used to promote the directed differentiation of stem cells into target cells by editing differentiation-related genes, so as to reduce the problems of non-differentiation, incomplete differentiation or differentiation into other cell types after stem cell transplantation.

Challenges and expectation in several strategies of constructing stem cell-based tissue-engineered tracheae

Directly inoculating stem cells into scaffolds (usually co-incubated stem cell suspensions with tracheal scaffolds to achieve adhesion and proliferation of stem cells on the scaffolds) is a simple and convenient strategy for stem cell delivery. And the tissue-engineered trachea constructed based on it has shown a good tracheal repair effect in some research, as shown in Table 1. However, it cannot be ignored that the efficiency of stem cell inoculation and the uniformity between batches in this strategy are both poor [237]. It cannot quickly deliver a large number of cells to the scaffold at one time because during the entire inoculation process, the stem cells need to first come into contact with the scaffold before gradually adhering to it, and eventually migrate and proliferate on the scaffold, which is time-consuming [238]. In addition, the efficiency of stem cell inoculation is also influenced by the microenvironment of the inoculation and culture, such as temperature, pH, oxygen concentration, and nutrient supply, which are crucial for cell adhesion and growth [239]. If it cannot be precisely regulated, it may lead to slow adhesion of stem cells, or even growth restriction or death.

The delivery of hydrogel loaded with stem cells to the tracheal scaffold can cellularize the tracheal scaffold more efficiently than direct inoculation of stem cells [203]. And hydrogels can provide a good growth environment for stem cells [240]. However, when high-density

stem cells are loaded into hydrogels, they may tend to cluster together, which can cause uneven cell density in the hydrogel and may lead to competition for nutrients among them, thereby affecting their normal growth and differentiation [241]. At present, hydrogels mainly used in the field of TTE are based on natural biomaterials, having good biodegradable properties and relatively weak mechanical properties [223]. When the hydrogels degrade too rapidly or cannot withstand the pressure from respiratory movements or surrounding tissues, their structure and integrity may be damaged, leading to repair failure due to the loss of supporting function before tissues fully repair. And even broken hydrogel fragments may fall into the tracheal cavity, causing airway obstruction and posing a threat to life. If the degradation is too slow, it can lead to the hydrogel remaining in the body for a long time, affecting the reconstruction of the natural structure of tissues and even possibly triggering an immune response. Therefore, in the preparation process of stem cell-loaded hydrogels, the loading method should be continuously optimized to achieve a uniform distribution of stem cells in the hydrogel. For example, one could try to introduce microfluidic technology to achieve a uniform distribution of stem cells in the hydrogel. The composition and cross-linking structure of hydrogels still need to be further explored to ensure the appropriate degradation rate of the hydrogel and the survival, proliferation, and differentiation of stem cells. Bioreactors that mimic the tracheal microenvironment can be designed and used to guide the optimization of hydrogel composition, porosity, and crosslinking to regulate the appropriate rate of hydrogel degradation and improve the growth space for stem cells, as well as the distribution, growth, and differentiation of stem cells. 3D bioprinting is a promising approach to stem cell-based TTE. Researchers can utilize this technology to develop a personalized biomimetic tracheal graft with hydrogels containing stem cells, resembling the structure and shape of the native trachea [222]. However, in the process of bio-printing, high shear force, dry humidity, temperature changes, and other factors may cause damage or death to stem cells [242, 243]. For optimal cell survival during the printing process, it is essential to precisely control the printing conditions and continuously optimize the viscosity of the hydrogel, printing rate, temperature, humidity, and so on. Furthermore, scaffolds of this type also require a certain swelling property to store nutrients that provide nutrients for the growth of stem cells until the formation of blood supply.

Stem cell sheets/stem cell spheres, as another form of cell delivery, are higher efficient than direct inoculation of stem cells, offering another promising way for researchers to reconstruct the complex tracheal structure [244]. However, both cell sheets and cell spheres are formed by the fusion and self-assembly of cells themselves, and they are usually prone to rupture by mechanical traction or extrusion [60, 245]. Therefore, maintaining structural integrity in the tracheal environment is something that needs to be focused on in the research. Furthermore, forming a hierarchical distribution and functional organization similar to that of a natural trachea through the application of stem cell sheets/spheres is a complex issue. In vitro induction of stem cell sheets/spheres to differentiate into tracheal component units (such as epithelial cell sheets/spheres, chondrocyte sheets/spheres, and smooth muscle cell sheets/spheres) can be used as raw materials for the reconstruction of each tracheal component [61, 118]. According to the structure of the natural gas tube, each tracheal component unit is modularly combined to prepare bionic tracheal grafts. Then the graft was cultivated in a bioreactor to promote the fusion and maturation of each component, thereby enhancing its integrity and mechanical properties. This is a possible method for reconstructing the multi-level structure of the trachea.

Stem cell-derived exosomes may be a potential therapeutic solution for promoting tissue regeneration while avoiding immune rejection and tumorigenicity of stem cells [246, 247]. However, the components and proportions of bioactive molecules carried by exosomes derived from stem cells from different sources are slightly different, and there are currently few studies in tracheal tissue engineering [248]. Therefore, research in this area needs to be expanded. Furthermore, it is necessary to verify the effects of stem cell-derived exosomes on different cell types of the trachea, providing a basis for future preparation of composite stem cell exosomes for tracheal repair. And prior to clinical application, further research is needed to demonstrate that the therapeutic efficacy of cell-free stents loaded with stem cell-derived exosomes is evident, safe and long-lasting [92].

Scaffold cartilage regeneration balance

To maintain airway patency following orthotopic transplantation, tissue-engineered trachea substitutes must exhibit long-lasting superior mechanical properties to prevent collapse and sustain a hollow structure. In addition, tissue-engineered tracheal substitutes are required to have good biocompatibility [249]. Therefore, polymers such as polylactic acid (PLA) and PCL, with good biocompatibility and good biomechanical properties in vivo, have been used to prepare tissue engineering tracheal scaffolds [8, 250–254]. However, these polymers are biodegradable. As the orthotopic transplantation time of this type of tissue engineering trachea scaffolds based on biodegradable polymer materials is gradually prolonged, the scaffold materials providing mechanical properties gradually degrade in vivo, eventually resulting in the destruction of the mechanical structure of the tracheal

scaffolds and the collapse of the tracheal grafts. Thus, it is particularly important for tracheal cartilage to regenerate and gradually replace degraded scaffolds to uphold the desirable biomechanical properties of tissue-engineered trachea. However, achieving a balance between tracheal scaffold absorption and cartilage regeneration poses severe challenges because it not only involves the degradation rate of tracheal scaffolds in the complex microenvironment in vivo and the time point of mechanical property collapse, but also involves the efficiency of stem cell cartilage differentiation and the duration for cartilage to grow to have sufficient mechanical properties on the tracheal scaffolds. The regulation of scaffold properties is the key to solving the problem of scaffold cartilage regeneration balance. Hence, it is necessary to develop suitable composite tracheal scaffolds according to the biodegradation rate, mechanical properties, and biocompatibility of different biomaterials [255]. The chondro-inducing factors such as TGF-β3 and BMP-2 can be added to the scaffold to promote differentiation of stem cells into cartilage and deposition of cartilage matrix, thereby enhancing the scaffold's ability to promote cartilage regeneration [256, 257]. The key focus should be on establishing in vitro models (such as bioreactors simulating the tracheal microenvironment) and in vivo models (such as animal in vivo embedding) to adjust the component ratio of the tracheal scaffold to optimize the degradation rate, mechanical properties and cartilage regeneration promotion efficiency of the scaffold, so as to guide the construction of tissue-engineered tracheal grafts that can achieve a balance of scaffold cartilage regeneration. Furthermore, in the future, the development of intelligent tracheal scaffolds that can adjust their mechanical properties and degradation rates based on environmental changes (such as airway stress, temperature, and pH) as well as cartilage regeneration conditions may be able to better adapt to the dynamic demands of tracheal tissue and maintain the required mechanical properties.

Overall, over the past few decades, the research in the field of stem cell-based TTE has undergone an evolution from the exploration of a single technology to the comprehensive application of multiple disciplines. Although current research has made certain progress and technologies are constantly integrating and maturing, there are still many problems to be solved, including the safety of stem cells, the differentiation efficiency of stem cells, the delivery strategy of stem cells, the selection and biocompatibility of scaffold materials, mechanical reconstruction, the standardized and mass-producible strategies, etc. for constructing tissue-engineered tracheas. Future research should focus on the optimization and clinical transformation of techniques and strategies for the construction of stem cell-based tissue-engineered tracheas, and pay more attention to the verification of safety and efficacy in clinical trials. Furthermore, multidisciplinary cooperation and cross-field innovation should be promoted to achieve standardized and large-scale preparation of tissue-engineered trachea, so as to provide safer and more efficient treatment for clinical practice.

Conclusion

Tissue-engineered trachea based on stem cells brings hope for the reconstruction of a long-segmented trachea. To advance research in this area, collaboration between countries/institutions has become the norm. The United States has the most production and influence in this field, and University College London has made an important contribution. In order to explore the construction of functional tissue-engineered trachea, researchers have used stem cells from various sources and have made many attempts, such as scaffolds directly seeded with stem cells, hydrogels encapsulating stem cells, stem cell sheets, and stem cell-derived exosomes. Tracheal cartilage reconstruction is the key to ensuring the long-term effect of tissue-engineered tracheal transplantation. The construction of chondrified tissue-engineered trachea based on the differentiation property and pro-regenerative ability of stem cells has been widely concerned. The development of hydrogel and 3D bioprinting is expected to realize the reconstruction of the anatomical structure and function of the trachea. Despite facing challenges at present, stem cell-based tissue-engineered trachea remains a promising clinical strategy.

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Author contributions

Jianwei Zhu: Conceptualization, Methodology, Formal analysis, Investigation, Writing—Original Draft; Min Bao: Methodology, Formal analysis, Investigation, Writing—Original Draft; Yibo Shan: Methodology, Formal analysis, Investigation, Writing—Original Draft; Yi Lu, Zhiming Shen, Wenxuan Chen, Hao Chen, and Lintao Guo: Investigation, Writing—Review & Editing; Hongcan Shi: Conceptualization, Resources, Writing—Review & Editing, Supervision, Project administration, Funding acquisition. All authors approved this manuscript.

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Data availability

Not applicable.

Declarations

Ethics approval and consent for publication

Not applicable.

Competing interest
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